Inventory Demand & Availability Dashboard

Dashboard Overview

This dashboard analyzes product demand, availability, and financial performance to help businesses optimize inventory management and reduce losses.

Problem Statement

This dashboard helps the business understand the gap between product demand and availability. It highlights supply shortages, financial losses, and inefficiencies in inventory management.

Through this dashboard, decision-makers can identify areas where demand exceeds supply and take necessary actions to improve inventory planning and reduce losses.

Since the **average demand is significantly higher than availability**, it indicates consistent supply shortages.

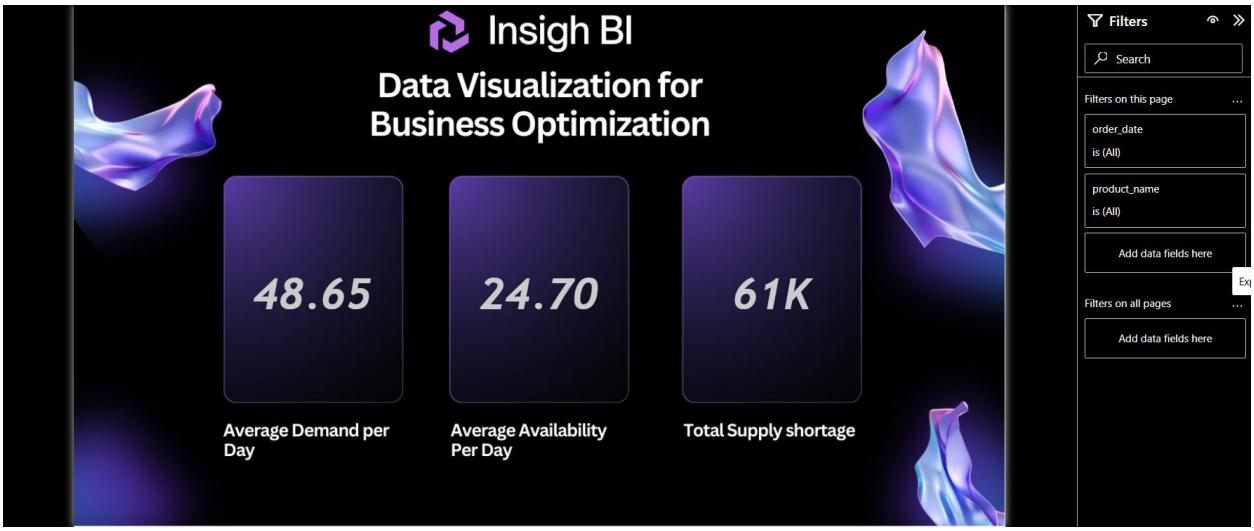
Also, the **total loss (8M) is much higher than total profit (301K)**, which shows major inefficiencies in inventory and supply chain management.

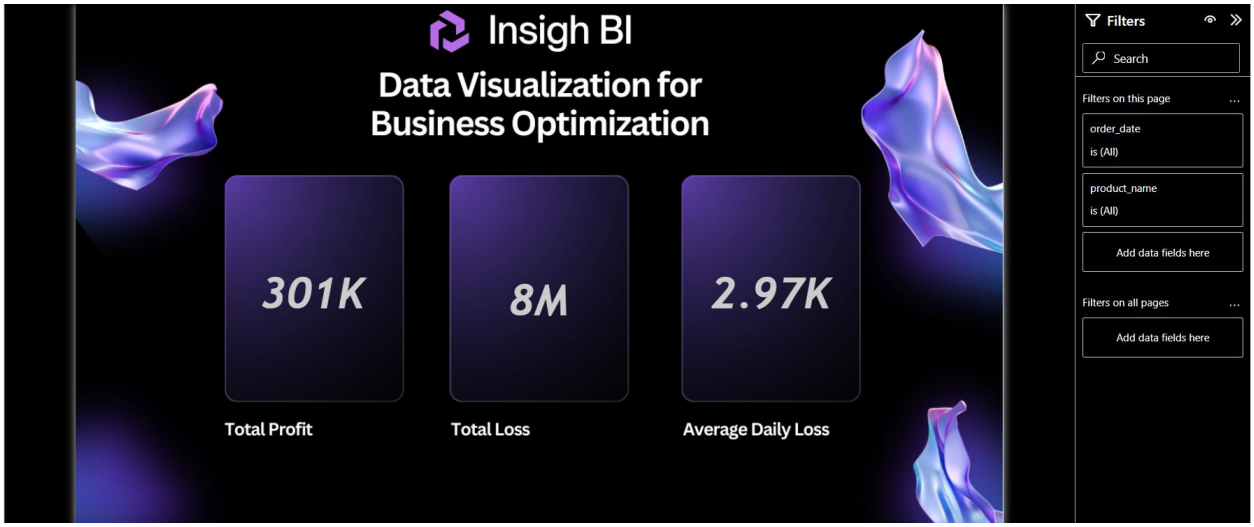
Steps Followed

- **Step 1:** Loaded dataset into Power BI Desktop from MySQL database.
- **Step 2:** Opened Power Query Editor and enabled:
    - Column Distribution
    - Column Quality
    - Column Profile
- **Step 3:** Switched profiling to entire dataset for accurate analysis.
- **Step 4:** Cleaned data and handled null/missing values where required.
- **Step 5:** Created necessary calculated measures using DAX.
- **Step 6:** Designed dashboard theme and layout for better visualization.
- **Step 7:** Created KPI cards for:
    - Average Demand

    - Average Availability
    - Supply Shortage
- **Step 8:** Added slicers (filters) for:
    - Order Date
    - Product Name
- **Step 9:** Created financial KPI visuals:
    - Total Profit
    - Total Loss
    - Average Daily Loss
- **Step 10:** Structured dashboard into two pages:
    - Demand vs Availability
    - Profit & Loss Analysis

Snapshot of Dashboard (Power BI Service)





Insights

A two-page dashboard was created on Power BI Desktop to analyze inventory and financial performance.

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[1] Demand vs Availability Analysis

- Average Demand per Day = **48.65**
- Average Availability per Day = **24.70**
- Total Supply Shortage = **61K**

-> Demand is almost **2x higher than availability**, indicating major supply shortages. -> This leads to missed sales opportunities and customer dissatisfaction.

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[2] Profit & Loss Analysis

- Total Profit = **301K**
- Total Loss = **8M**
- Average Daily Loss = **2.97K**

-> Loss is significantly higher than profit. -> Indicates poor inventory planning and inefficiencies in operations.

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[3] Key Observations

- High mismatch between demand and supply
- Continuous supply shortages across products
- Financial losses dominating profits
- Inefficient inventory distribution

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[4] Business Insights

- Inventory levels are not aligned with demand patterns
- High-demand products are understocked
- Losses may be due to:
    - Stockouts (lost sales)
    - Overstocking (holding cost)

Recommendations

- Implement **demand forecasting models**
- Maintain optimal **safety stock levels**
- Improve **inventory replenishment strategies**
- Focus on high-demand products
- Use dashboards for real-time monitoring

Conclusion

This project demonstrates how data analytics can uncover inefficiencies in inventory management. By using Power BI dashboards, businesses can improve supply chain performance, reduce losses, and make data-driven decisions.

Project Files

- Power BI Dashboard (.pbix)

- MySQL Dataset
- MySQL Queries